HBSC: Data Overview and Prep

Updated: 2026-08-02

Dataset Overview

File has the following rows and columns:

```
## [1] 244097    120
```

Pretty much all columns have missing values so need to address later. Though it seems much of the variables I will care about don't have much.

```
##             variable na_count na_pct
## 1             agesex   228488   93.6
## 2     contraceptpill   228448   93.6
## 3   contraceptcondom   228070   93.4
## 4        employnotfa   224524   92.0
## 5        employnotmo   199890   81.9
## 6             hadsex   162986   66.8
## 7       cannabis30d_2   144442   59.2
## 8       cannabisltm_2   142745   58.5
## 9       countrybornfa    71901   29.5
## 10      countrybornmo    70659   28.9
## 11        countryborn    69118   28.3
## 12        talkstepmo    69006   28.3
## 13        talkstepfa    67025   27.5
## 14             grade    64830   26.6
## 15       oweight_who    54573   22.4
## 16             IOTF4    54541   22.3
## 17              MBMI    53311   21.8
## 18        bodyweight    41128   16.8
## 19        bodyheight    40036   16.4
## 20        emcsocmed8    32878   13.5
## 21        emcsocmed9    32855   13.5
## 22        emcsocmed7    32597   13.4
## 23        emcsocmed6    32544   13.3
## 24        emcsocmed5    32443   13.3
## 25        emcsocmed4    32428   13.3
## 26        emcsocmed3    32009   13.1
## 27        emcsocmed1    31973   13.1
## 28        emcsocmed2    31659   13.0
## 29       emconlpref3    31097   12.7
## 30       emconlpref2    30586   12.5
## 31       emconlpref1    29813   12.2
## 32       emconlfreq2    29552   12.1
## 33       emconlfreq4    29003   11.9
```

```
## 34       emconlfreq3    28973   11.9
## 35       emconlfreq1    26720   10.9
## 36          employmo    24135    9.9
## 37          employfa    23999    9.8
## 38     bulliedothers    21579    8.8
## 39               id3    20807    8.5
## 40         smok30d_2    20687    8.5
## 41           smokltm    20451    8.4
## 42          drunk30d    20415    8.4
## 43            famdec    20237    8.3
## 44            famsup    20091    8.2
## 45           famtalk    20019    8.2
## 46          alc30d_2    19326    7.9
## 47          drunkltm    19266    7.9
## 48           famhelp    18896    7.7
## 49       teachercare    18340    7.5
## 50            alcltm    18007    7.4
## 51      teachertrust    17839    7.3
## 52        talkfather    17491    7.2
## 53       breakfastwe    17468    7.2
## 54        studaccept    17444    7.1
## 55       beenbullied    16962    6.9
## 56     teacheraccept    16764    6.9
## 57       studhelpful    16746    6.9
## 58      studtogether    16351    6.7
## 59        talkmother    15521    6.4
## 60      cbeenbullied    15084    6.2
## 61    cbulliedothers    14934    6.1
## 62             fmeal    14738    6.0
## 63             IRFAS    12599    5.2
## 64      IRRELFAS_LMH    12599    5.2
## 65       fasdishwash    12124    5.0
## 66         thinkbody    11657    4.8
## 67        elsehome1_2    11124    4.6
## 68       fosterhome1    10938    4.5
## 69    sleepdificulty    10814    4.4
## 70       friendshare    10813    4.4
## 71        friendtalk    10808    4.4
## 72       breakfastwd    10748    4.4
## 73       fasbathroom    10743    4.4
## 74       stepfahome1    10451    4.3
## 75       stepmohome1    10449    4.3
## 76     friendcounton    10374    4.2
## 77       motherhome1    10235    4.2
## 78       fatherhome1    10163    4.2
## 79        friendhelp     9744    4.0
## 80        injured12m     8974    3.7
## 81          fight12m     8762    3.6
## 82             dizzy     8566    3.5
## 83           toothbr     7910    3.2
## 84       fasholidays     7727    3.2
## 85           feellow     6696    2.7
## 86        fasbedroom     6672    2.7
## 87          backache     6543    2.7
```

```
## 88          irritable     6539    2.7
## 89            nervous     6525    2.7
## 90          physact60     6215    2.5
## 91          fasfamcar     6195    2.5
## 92      vegetables_2     6021    2.5
## 93          sweets_2     6004    2.5
## 94      softdrinks_2     5906    2.4
## 95        stomachache     5679    2.3
## 96       fascomputers     5527    2.3
## 97     schoolpressure     5486    2.2
## 98          fruits_2     5294    2.2
## 99         likeschool     5109    2.1
## 100           timeexe     4799    2.0
## 101           lifesat     4512    1.8
## 102          headache     4505    1.8
## 103            health     2577    1.1
## 104               age     1516    0.6
## 105            agecat     1516    0.6
## 106         yearbirth     1516    0.6
## 107        monthbirth     1106    0.5
## 108             month       18    0.0
## 109              HBSC        0    0.0
## 110         seqno_int        0    0.0
## 111           cluster        0    0.0
## 112         countryno        0    0.0
## 113            region        0    0.0
## 114               id1        0    0.0
## 115               id2        0    0.0
## 116               id4        0    0.0
## 117            weight        0    0.0
## 118               adm        0    0.0
## 119              year        0    0.0
## 120               sex        0    0.0
```

seqno_int

This is the unique identifier of each record and I confirmed it's unique.

```r
length(unique(dat$seqno_int)) == nrow(dat)
```

```
## [1] TRUE
```

countryno

Number of countries in dataset is:

```r
length(unique(dat$countryno))
```

```
## [1] 47
```

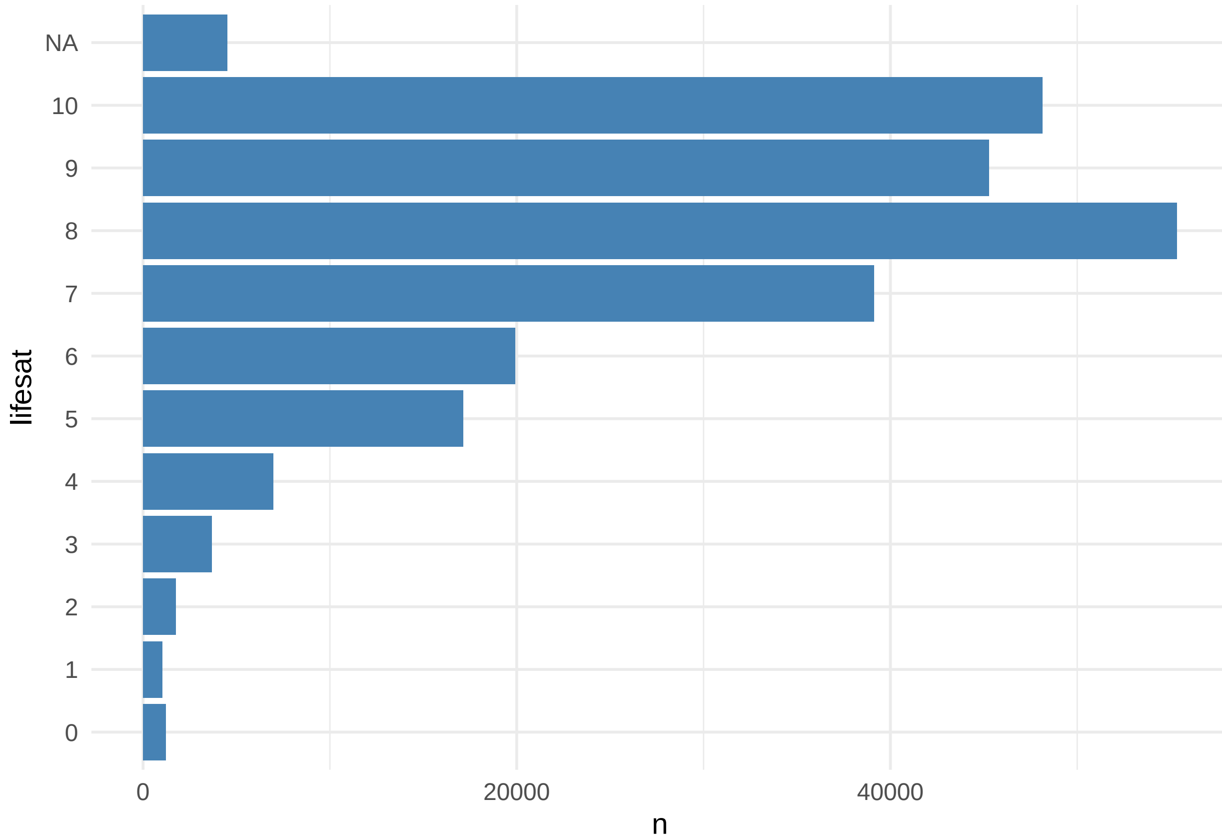
And this is the count of survey responses by country:

```
## # A tibble: 46 x 3
##    country_name            n   pct
##    <chr>               <int> <dbl>
##  1 Wales               15951   6.5
##  2 Canada              12950   5.3
##  3 Czech Republic      11564   4.7
##  4 France               9170   3.8
##  5 <NA>                 8411   3.4
##  6 Israel               7712   3.2
##  7 Switzerland          7510   3.1
##  8 Iceland              6996   2.9
##  9 Ukraine              6660   2.7
## 10 Portugal             6126   2.5
## 11 Turkey               5848   2.4
## 12 Slovenia             5667   2.3
## 13 Belgium (French)     5578   2.3
## 14 Poland               5224   2.1
## 15 Croatia              5169   2.1
## 16 Scotland             5021   2.1
## 17 Kazakhstan           4868   2
## 18 Slovakia             4785   2
## 19 Estonia              4725   1.9
## 20 Armenia              4717   1.9
## 21 Netherlands          4698   1.9
## 22 Republic of Moldova  4686   1.9
## 23 Macedonia            4658   1.9
## 24 Azerbaijan           4586   1.9
## 25 Romania              4567   1.9
## 26 Latvia               4412   1.8
## 27 Germany              4347   1.8
## 28 Belgium (Flemish)    4333   1.8
## 29 Spain                4320   1.8
## 30 Russia               4281   1.8
## 31 Georgia              4242   1.7
## 32 Sweden               4185   1.7
## 33 Italy                4144   1.7
## 34 Austria              4129   1.7
## 35 Luxembourg           4070   1.7
## 36 Serbia               3933   1.6
## 37 Ireland              3833   1.6
## 38 Lithuania            3797   1.6
## 39 Hungary              3789   1.6
## 40 England              3397   1.4
## 41 Denmark              3181   1.3
## 42 Finland              3146   1.3
## 43 Norway               3127   1.3
## 44 Malta                2576   1.1
## 45 Albania              1765   0.7
## 46 Greenland            1243   0.5
```

Variables to Predict

lifesat

Q: Here is a picture of a ladder. The top of the ladder "10" is the best possible life for you and the bottom "0" is the worst possible life for you. In general, where on the ladder do you feel you stand at the moment?
A: 0-10



lifesat is dichotomized for modeling as lifesat_low:

```
##
##              0     1  <NA>
##    0         0  1218     0
##    1         0  1041     0
##    2         0  1748     0
##    3         0  3691     0
##    4         0  6961     0
##    5         0 17137     0
##    6     19921     0     0
##    7     39120     0     0
##    8     55325     0     0
##    9     45271     0     0
##    10    48152     0     0
##    <NA>      0     0  4512
```

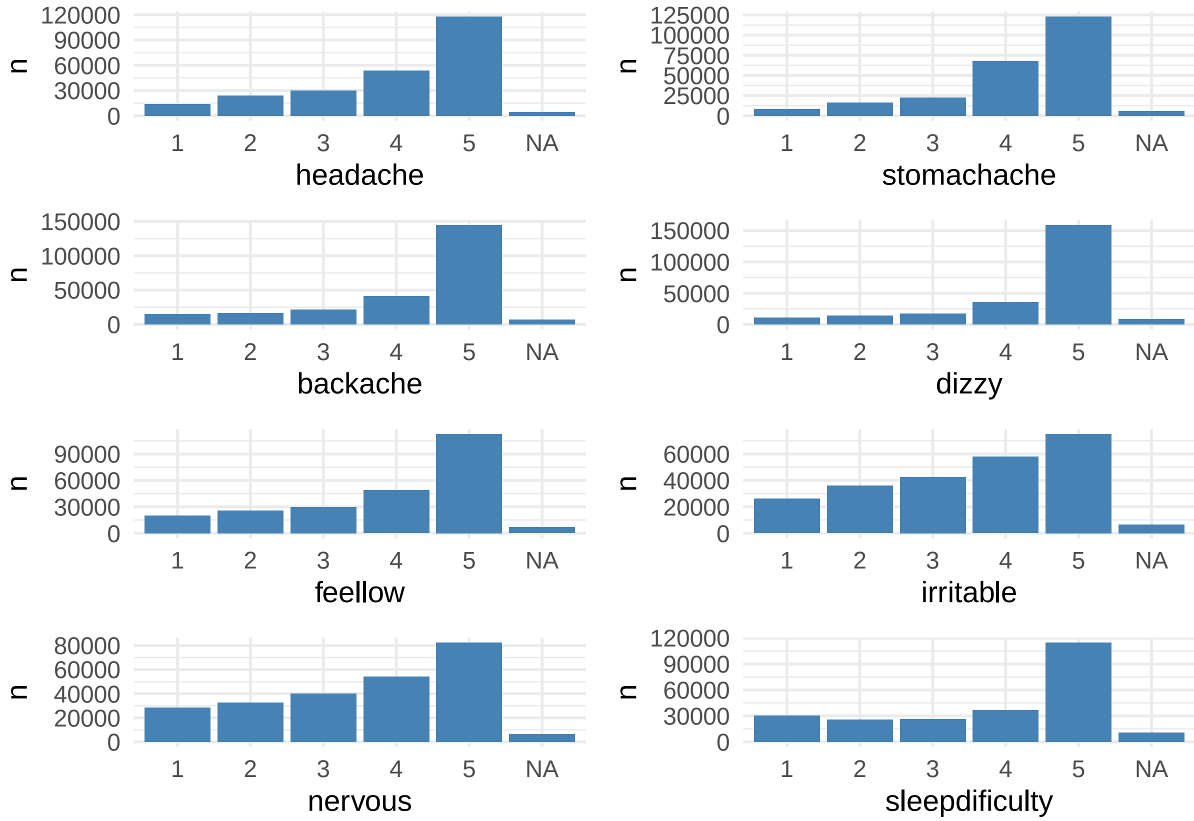
Ultimately the distribution is:

```
## # A tibble: 3 x 3
##   lifesat_low      n    pct
##         <dbl>  <int>  <dbl>
## 1           0 207789 0.851
## 2           1  31796 0.130
## 3          NA   4512 0.0185
```

(mental) health complaints

There are 8 variables for this that reply to a question like: In the last 6 months: how often have you had the following....? And the answers goes from 1 About every day to 5 Rarely or never. The 8 variables are: headache, stomachache, backache, dizzy, feellow, irritable, nervous, sleepdificulty.

Their distributions look like this:



These variables are then recoded to a binary where it flags if the complaint occurs more than once per week (1 or 2).

The recoded variables look like this:

```
##
##
## ### headache x headache_frequent
##
##              0      1   <NA>
##   1          0  13883      0
##   2          0  24257      0
##   3      30076      0      0
```

```
##   4     53676      0      0
##   5    117700      0      0
##   <NA>      0      0   4505
##
##
## ### stomachache x stomachache_frequent
##
##             0      1   <NA>
##   1         0   8415      0
##   2         0  16223      0
##   3     22895      0      0
##   4     68018      0      0
##   5    122867      0      0
##   <NA>      0      0   5679
##
##
## ### backache x backache_frequent
##
##             0      1   <NA>
##   1         0  14872      0
##   2         0  16137      0
##   3     21410      0      0
##   4     40910      0      0
##   5    144225      0      0
##   <NA>      0      0   6543
##
##
## ### dizzy x dizzy_frequent
##
##             0      1   <NA>
##   1         0  10461      0
##   2         0  14272      0
##   3     17261      0      0
##   4     35189      0      0
##   5    158348      0      0
##   <NA>      0      0   8566
##
##
## ### feellow x feellow_frequent
##
##             0      1   <NA>
##   1         0  20224      0
##   2         0  25739      0
##   3     29784      0      0
##   4     48898      0      0
##   5    112756      0      0
##   <NA>      0      0   6696
##
##
## ### irritable x irritable_frequent
##
##            0     1  <NA>
##   1        0 26053     0
##   2        0 35879     0
```

```
##   3    42648     0     0
##   4    57922     0     0
##   5    75056     0     0
##   <NA>     0     0  6539
##
##
## ### nervous x nervous_frequent
##
##            0     1  <NA>
##   1        0 28306     0
##   2        0 32770     0
##   3    40064     0     0
##   4    54109     0     0
##   5    82323     0     0
##   <NA>     0     0  6525
##
##
## ### sleepdificulty x sleepdificulty_frequent
##
##             0      1   <NA>
##   1         0  30501      0
##   2         0  25543      0
##   3     26053      0      0
##   4     36570      0      0
##   5    114616      0      0
##   <NA>      0      0  10814
```

The frequency of complaints is then added to calculate one sum for physical and one for mental. If the sum is 2 or more then it is determined that there is a health issue. Most papers do it for psycosomatic (all 8 vars) but I will focus on the 4 mental ones. Still, I include both vars in case.

```
##
##
## ### mental_complaintsY x mental_complaints_sum

##
##             0      1   <NA>
##   0    121573      0      0
##   1     45509      0      0
##   2         0  29991      0
##   3         0  21172      0
##   4         0  12970      0
##   <NA>      0      0  12882

##
##
## ### (health_complaintsY x health_complaints_sum

##
##             0      1   <NA>
##   0    103678      0      0
##   1     41923      0      0
##   2         0  29221      0
```

```
##   3          0  21360      0
##   4          0  14640      0
##   5          0   8824      0
##   6          0   5202      0
##   7          0   2839      0
##   8          0   2219      0
##   <NA>       0      0  14191
```

mental_complaintsY

Ultimately our variable to predict will be mental_complaintsY; at ~26%.

```
## # A tibble: 3 x 3
##   mental_complaintsY      n    pct
##                <dbl>  <int>  <dbl>
## 1                  0 167082 0.684
## 2                  1  64133 0.263
## 3                 NA  12882 0.0528
```

The Predictors

Problematic social media use (PMSU)

PMSU are the main variables of interest. There are 9 of them: emcsocmed1 to emcsocmed9. The prequestion for all of them is: We are interested in your experiences with social media. The term social media refers to social network sites (e.g. Facebook, [add other local examples]) and instant messengers (e.g. [insert local examples], WhatsApp, Snapchat, Facebook messenger). During the past year, have you. . . . Please tick one box for each line.

The possible answers are 1 (no) and 2 (yes). Also those with 99 mean Missing due to skip pattern.

The topic of each question is:

item	topic
emcsocmed1	Social media: I can't think of anything else
emcsocmed2	Social media: Dissatisfied cause want to spend more time
emcsocmed3	Social media: Felt bad cause want to play
emcsocmed4	Social media: Failed to spend less time
emcsocmed5	Social media: Neglected other activities
emcsocmed6	Social media: Had arguments because of use
emcsocmed7	Social media: Lied about amount of time spent
emcsocmed8	Social media: Use it to escape from negative feelings
emcsocmed9	Social media: Conflict with family because of use

Since I need to get a sum of them I will recode them from 1/2 to 0/1.These are the distributions of the answers. I put the 99 as NA - is that correct?

The recoding looks like this:

```
## ## Table for emcsocmed1
##              0      1   <NA>
##   1    156598      0      0
```

```
##   2         0  44267      0
##   99        0      0  11259
##   <NA>      0      0  31973
##
##
## ## Table for emcsocmed2
##             0      1   <NA>
##   1    163166      0      0
##   2         0  38013      0
##   99        0      0  11259
##   <NA>      0      0  31659
##
##
## ## Table for emcsocmed3
##             0      1   <NA>
##   1    157891      0      0
##   2         0  42938      0
##   99        0      0  11259
##   <NA>      0      0  32009
##
##
## ## Table for emcsocmed4
##             0      1   <NA>
##   1    138631      0      0
##   2         0  61779      0
##   99        0      0  11259
##   <NA>      0      0  32428
##
##
## ## Table for emcsocmed5
##             0      1   <NA>
##   1    168967      0      0
##   2         0  31428      0
##   99        0      0  11259
##   <NA>      0      0  32443
##
##
## ## Table for emcsocmed6
##             0      1   <NA>
##   1    162482      0      0
##   2         0  37812      0
##   99        0      0  11259
##   <NA>      0      0  32544
##
##
## ## Table for emcsocmed7
##             0      1   <NA>
##   1    171121      0      0
##   2         0  29120      0
##   99        0      0  11259
##   <NA>      0      0  32597
##
##
## ## Table for emcsocmed8
```

```
##             0      1   <NA>
##   1    137767      0      0
##   2         0  62193      0
##   99        0      0  11259
##   <NA>      0      0  32878
##
##
## ## Table for emcsocmed9
##             0      1   <NA>
##   1    171010      0      0
##   2         0  28973      0
##   99        0      0  11259
##   <NA>      0      0  32855
```

And this gives an idea of how correlated they are with each other.

Is this right? Too correlated? Or it's done different? It has to be done another way?

```
##            emcsocmed1 emcsocmed2 emcsocmed3 emcsocmed4 emcsocmed5 emcsocmed6
## emcsocmed1     1.0000     0.9998     0.9998     0.9997     0.9998     0.9998
## emcsocmed2     0.9998     1.0000     0.9998     0.9997     0.9998     0.9998
## emcsocmed3     0.9998     0.9998     1.0000     0.9997     0.9998     0.9998
## emcsocmed4     0.9997     0.9997     0.9997     1.0000     0.9997     0.9997
## emcsocmed5     0.9998     0.9998     0.9998     0.9997     1.0000     0.9998
## emcsocmed6     0.9998     0.9998     0.9998     0.9997     0.9998     1.0000
## emcsocmed7     0.9998     0.9998     0.9998     0.9998     0.9998     0.9998
## emcsocmed8     0.9997     0.9997     0.9997     0.9997     0.9997     0.9997
## emcsocmed9     0.9998     0.9998     0.9998     0.9998     0.9998     0.9998
##            emcsocmed7 emcsocmed8 emcsocmed9
## emcsocmed1     0.9998     0.9997     0.9998
## emcsocmed2     0.9998     0.9997     0.9998
## emcsocmed3     0.9998     0.9997     0.9998
## emcsocmed4     0.9998     0.9997     0.9998
## emcsocmed5     0.9998     0.9997     0.9998
## emcsocmed6     0.9998     0.9997     0.9998
## emcsocmed7     1.0000     0.9998     0.9999
## emcsocmed8     0.9998     1.0000     0.9998
## emcsocmed9     0.9999     0.9998     1.0000
```

PMSU variables are then combined so output is represented in one variable, instead of 9. I will do 2 versions: a) Yes/No where Y(6-9) and N(0-5) b) H/M/L where H(6-9), M(2-5) and L(0-1)

The distribution of PMSU categorized is:

```
## ## Table for emcsocmed_sum x pmsu_yn
```

```
##
##            0     1  <NA>
##   0    72474     0     0
##   1    36277     0     0
##   2    26066     0     0
##   3    19012     0     0
##   4    14908     0     0
##   5    10306     0     0
```

```
##   6          0  5905      0
##   7          0  3382      0
##   8          0  1878      0
##   9          0  2979      0
##   <NA>       0     0 50910
```

```
## ## Table for emcsocmed_sum x pmsu_lmh
```

```
##
##          High   Low   Med  <NA>
##   0         0 72474     0     0
##   1         0 36277     0     0
##   2         0     0 26066     0
##   3         0     0 19012     0
##   4         0     0 14908     0
##   5         0     0 10306     0
##   6      5905     0     0     0
##   7      3382     0     0     0
##   8      1878     0     0     0
##   9      2979     0     0     0
##   <NA>      0     0     0 50910
```

```
## ## Table for pmsu_yn x pmsu_lmh
```

```
##
##              0      1   <NA>
##   High       0  14144      0
##   Low   108751      0      0
##   Med    70292      0      0
##   <NA>       0      0  50910
```

Family Support

There are 4 of them. The question starts with: We are interested in how you feel about the following statements. Please show how much you agree or disagree with each one. Please tick one box for each line.

The options to answer go from 1 Very strongly disagree to 7 Very strongly agree. The 4 questions are:

item	topic
famhelp	Family tries to help
famsup	Get emotional help from family
famtalk	Talk about problems with family
famclose	Feel close to family

These variables are then combined to create a categorical predictor. For now, I will keep two options like I did for PMSU but will select only one when modeling. These are:
a) H/L based on avg of 4 vars. L if [1,5.5) and H if [5.5, 7]
b) H/M/L based on sum of 4 vars. L if 4-11, M if 12-19, H if 20-28.

Their counts after recoding are:

Table 3: family_support_high

Var1	Freq	pct
0	65676	0.269
1	156068	0.639
NA	22353	0.092

Table 4: family_support_hml

Var1	Freq	pct
High	169660	0.695
Low	23345	0.096
Med	28739	0.118
NA	22353	0.092

Frieds (Peer) Support

There are 4 friends support questions. The questions start with: We are interested in how you feel about the following statements. Please show how much you agree or disagree with each one. Please tick one box for each line.

The answers range from 1 Very strongly disagree to 7 Very strongly agree.

item	topic
friendhelp	My friends really try to help me
friendcounton	I can count on my friends when things go wrong
friendshare	I have friends with whom I can share my joys and sorrows
friendtalk	I can talk about my problems with my friends

These variables are then combined to create a categorical predictor. For now, I will keep two options like I did for family but will select only one when modeling. These are:
a) H/L based on avg of 4 vars. L if [1,5.5) and H if [5.5, 7]
b) H/M/L based on sum of 4 vars. L if 4-11, M if 12-19, H if 20-28.

Their counts after recoding are:

Table 6: friends_support_high

Var1	Freq	pct
0	95821	0.393
1	135380	0.555
NA	12896	0.053

Table 7: friends_support_hml

Var1	Freq	pct
High	155201	0.636
Low	29537	0.121

Var1	Freq	pct
Med	46463	0.190
NA	12896	0.053

IRRELFAS_LMH: Relative family affluence categorical

Derived from IRFAS. So check list of fas-variables in the dataset. It is a cut based on percentiles so I recode to friendly. 1 Lowest 20 pct
2 Medium 60 pct
3 Highest 20 pct

Table 8: IRRELFAS_LMH x IRRELFAS_LMH_r

	High20	Low20	Med60	NA
1	0	45818	0	0
2	0	0	142148	0
3	43532	0	0	0
NA	0	0	0	12599

talkfather and talkmother

Two questions measure parent communication. The pre-question is: How easy is it for you to talk to the following persons about things that really bother you? Please tick one box for each line

Answers go from 1 Very easy to 5 Don't have or see.

These are dichotomized for each parent as Yes if [1,2] and No if [3-5].

```
## ## Table for talkfatherYes x talkfather
```

```
##
##              0     1  <NA>
##   1          0 73777     0
##   2          0 78346     0
##   3      40289     0     0
##   4      19010     0     0
##   5      15184     0     0
##   <NA>       0     0 17491
```

```
## ## Table for talkmotherYes x talkmother
```

```
##
##               0      1   <NA>
##   1           0 115899      0
##   2           0  74913      0
##   3       24042      0      0
##   4        9276      0      0
##   5        4446      0      0
##   <NA>        0      0  15521
```

I was going to combine to a "number of parents one can talk" but I'll leave for now and will try as interactions later.

bullying victim

There are 2 variables that measure if kid has been bullied. a) beenbullied: Been bullied past months and b) cbeenbullied: Been cyber bullied. Answers range from 1 Haven't to 5 Several times a week.

These are dichotomized as Yes if [2-5] and No if [1].

```
## ## Table for beenbullied x beenbulliedYes
```

```
##
##              0      1   <NA>
##   1    166254      0      0
##   2         0  36624      0
##   3         0   9754      0
##   4         0   6179      0
##   5         0   8324      0
##   <NA>      0      0  16962
```

```
## ## Table for cbeenbullied x cbeenbulliedYes
```

```
##
##              0      1   <NA>
##   1    198900      0      0
##   2         0  20426      0
##   3         0   4625      0
##   4         0   2274      0
##   5         0   2788      0
##   <NA>      0      0  15084
```

age and sex(gender)

There are 3 age groups and 2 genders. I recode to friendly/actual value and they look like this.

```
## ## Table for agecat x age_r
```

```
##
##            11    13    15  <NA>
##   1    81834     0     0     0
##   2        0 83749     0     0
##   3        0     0 76998     0
##   <NA>     0     0     0  1516
```

```
## ## Table for sex x sex_r
```

```
##
##            Boy   Girl   <NA>
##   1    120350      0      0
##   2         0 123747      0
##   <NA>      0      0      0
```

```
## ## Table for age_r x sex_r
```

```
##
##          Boy  Girl  <NA>
##   11   40476 41358     0
##   13   41612 42137     0
##   15   37394 39604     0
##   <NA>   868   648     0
```

Dataset for Modeling

Only the relevant and prepared variables are exported to a csv so I can start the modeling process with those. File goes to "data/processed/" and includes date.